

GitHub is the central repository where the code is maintained. Bitbucket and Nexus are the other repository tools.

Database management: This helps in managing revisions of database schema scripts. Liquibase is a widely used open source database solution that supports various databases.

Continuous monitoring: Continuous monitoring across all phases of application development, testing and deployment is crucial for a successful DevSecOps implementation. Improving service quality by monitoring application performance and log management solves the problem of aggregating, storing, and analysing all logs in one place.

Splunk, ELK Stack, Nagios, Sensu, and NewRelic are some of the popular tools for monitoring.

Open source tools for DevSecOps adoption in the cloud

Open source DevSecOps tools for the cloud are designed and developed using open source technologies to fulfil the DevSecOps toolchain capabilities. These are:

- Portfolio management tools, which provide transparency to stakeholders and participants
- Collaboration tools to help teams work together, anywhere and anytime
- Source control tools, which are the single source of truth
- Issue tracking tools to increase responsiveness and visibility
- Configuration management tools that enforce the desired state
- Continuous integration tools
- Binary repositories that manage builds, releases and dependencies
- Monitoring tools, which ensure service uptime and optimal performance
- Automated test tools for higher quality
- Deployment tools to improve time to market
- Security tools to provide security in the DevOps life cycle, covering interactive application security and testing, runtime application and self-protection, and cloud security

The following are some of the key open source DevSecOps tools that can be used in the cloud.

Ansible: Red Hat owns Ansible. This tool automates various common tasks related to IT operations such as application deployment, configuration management and cloud provisioning. It integrates with numerous DevOps tools including Jenkins, JIRA, Git, and many others. The free open source version of Ansible is available on GitHub.

Chef: Chef is an open source automation platform that transforms infrastructure into code. It operates in the cloud, on-premises, or in a hybrid environment. The Chef development kit provides the tools to develop and test infrastructure automation code from a local workstation prior to deploying changes into production.

Docker: Docker is software used for OS level virtualisation. It is used to create, deploy and run application packages called containers. Containers allow the developer to package an application with all the parts it needs, such as libraries and other dependencies, and ship it as one package. Docker is lightweight, open and secure.

Docker has two parts. Docker Engine is a tool responsible for creating and running Docker containers. Docker Hub is a service application based on the cloud, which covers the concept of application sharing and workflow automation.

GitHub: This is a collaborative code review tool supporting around 200 software languages. It also supports all the version control features of check-in, commits, branches, merging, labels, task management, wikis, push and pull to/from GitHub, etc. Git fits in very well as a popular and distributed version control system for teams located at different geographical locations.

Many DevSecOps teams use it for managing the source code of their applications. It consists of plugins that can link with Jenkins to facilitate deployment and integration.

Hudson: This continuous integration tool is developed in Java and runs on a VMware host or cloud. It is used for managing, monitoring, continuous testing and integration. It supports various systems for management of source code, application servers, code analysis tools, testing frameworks, and build tools. It has real-time notifications of test failures, change set support, and an easy installation and configuration process.

Jenkins: Jenkins is a cloud based continuous integration tool that helps to automate the activities of build, code analysis and storing of artefacts. These activities are triggered once a developer or the team commits the code to the version control repository.

Jenkins has many plugins and works as a CI tool for various different technologies like C/C++, Java/J2EE, .NET, Angular JS, etc. It also provides plugins to integrate with SonarQube for code review, JFrog Artifactory for storing binary artifacts, testing tools like Selenium, etc, as a part of the automation process.

Jenkins helps to automate deployments to app servers like Tomcat, JBoss, Weblogic through plugins, and also to container platforms like Docker.